Mat 1575 HW 5 (§1.6)

In the following exercises, compute each indefinite integral.

NOTE: $2^x = e^{\ln 2^x} = e^{x\ln 2}$

320. $\int e^{2x}\,dx$

$u=2x,\ du=2dx$

$\frac{du}{2}=dx$

$=\int e^u \frac{du}{2}$

$=\frac{e^u}{2}+C$

$=\frac{e^{2x}}{2}+C$

321. $\int e^{-3x}\,dx$

$u=-3x,\ du=-3dx$

$\frac{du}{-3}=dx$

$=\int e^u \frac{du}{-3}$

$=\frac{e^u}{-3}+C$

$=\frac{e^{-3x}}{-3}+C$

322. $\int 2^x\,dx$

$=\int e^{\ln 2^x}\,dx$

$=\int e^{x\ln 2}\,dx$

$=\frac{e^{x\ln 2}}{\ln 2}+C$

$=\frac{2^x}{\ln 2}+C$

325. $\int \frac{2}{x}\,dx$

$=2\ln|x|+C$

327. $\int \frac{1}{\sqrt{x}}\,dx$

$=\frac{1}{2}x^{\frac{1}{2}}+C$

In the following exercises, find each indefinite integral by using appropriate substitutions.

328. $\int \frac{\ln x}{x}\,dx = \int u\,du = \frac{u^2}{2}+C = \frac{(\ln x)^2}{2}+C$

$u=\ln x,\ du=\frac{dx}{x}$

330. $\int \frac{dx}{x\ln x}\ (x>1) \Rightarrow \int \frac{dx}{x\ln x} = \int \frac{du}{u} = \ln|u|+C = \ln|\ln x|+C$

$u=\ln x,\ du=\frac{dx}{x}$

332. $\int \tan\theta\,d\theta = \int \frac{\sin\theta}{\cos\theta}\,d\theta = \int \frac{-du}{u} = -\ln|u|+C = -\ln|\cos\theta|+C$

$u=\cos\theta,\ du=-\sin\theta\,d\theta,\ -du=\sin\theta\,d\theta$

335. $\int \ln(\cos x)\tan x\,dx = \int u\cdot(-du) = \int -u\,du = -\frac{u^2}{2}+C$

$u=\ln(\cos(x)),\ du=\frac{-\sin(x)}{\cos(x)}dx=-\tan(x)\,dx$

$\Rightarrow -du=\tan(x)\,dx$

$=-\frac{(\ln|\cos(x)|)^2}{2}+C$

337. $\int x^2 e^{-x^3}\,dx$ $= \int e^u \frac{du}{-3} = \frac{e^u}{-3} + C = \frac{e^{x^3}}{-3} + C$

$u = -x^3,\ du = -3x^2\,dx,\ \frac{du}{-3} = x^2\,dx$

338. $\int e^{\sin x}\cos x\,dx$ $= \int e^u\,du = e^u + C = e^{\sin(x)} + C$

$u = \sin(x).\ du = \cos(x)\,dx$

In the following exercises, evaluate the definite integral.

355. $\int_1^2 \frac{1+2x+x^2}{3x+3x^2+x^3}\,dx$

$u = 3x+3x^2+x^3$

$du = 3+6x+3x^2\,dx$

$\frac{du}{3} = 1+2x+x^2\,dx$

$= \int_{u(1)}^{u(2)} \frac{1}{u}\frac{du}{3}$

$= \int_7^{26} \frac{du}{3u}$

$= \frac{\ln|u|}{3}\Big|_7^{26}$

$= \frac{1}{3}(\ln 26 - \ln 7)$

356. $\int_0^{\pi/4} \tan x\,dx$

$u = \cos(x)$

$du = -\sin(x)\,dx$

$= \int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos(x)}\,dx = \int_1^{\frac{\sqrt{2}}{2}} \frac{-du}{u}$

$= \int_{\frac{\sqrt{2}}{2}}^1 \frac{du}{u} = \ln|u|\Big|_{\frac{\sqrt{2}}{2}}^1$

$= \ln 1 - \ln\frac{\sqrt{2}}{2}$

$= -\ln\frac{\sqrt{2}}{2}$

357. $\int_0^{\pi/3} \frac{\sin x - \cos x}{\sin x + \cos x}\,dx$

$u = \sin(x)+\cos(x)$

$du = (\cos(x)-\sin(x))\,dx$

$-du = (-\cos(x)+\sin(x))\,dx$

$= \int_1^{\frac{1+\sqrt{3}}{2}} \frac{-du}{u}$

$= \int_{\frac{1+\sqrt{3}}{2}}^1 \frac{du}{u}$

$= \ln|u|\Big|_{\frac{1+\sqrt{3}}{2}}^1$

$= \ln 1 - \ln\left(\frac{1+\sqrt{3}}{2}\right) = -\ln\left(\frac{1+\sqrt{3}}{2}\right)$

358. $\int_{\pi/6}^{\pi/2} \csc x\,dx$

$= \left[\ln|\csc(x)-\cot(x)|\right]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$

$= \ln\left|\csc\left(\frac{\pi}{2}\right)-\cot\left(\frac{\pi}{2}\right)\right| - \ln\left|\csc\left(\frac{\pi}{6}\right)-\cot\left(\frac{\pi}{6}\right)\right|$

$= \ln\left|\frac{1}{\sin\left(\frac{\pi}{2}\right)} - \frac{\cos\left(\frac{\pi}{2}\right)}{\sin\left(\frac{\pi}{2}\right)}\right| - \ln\left|\frac{1}{\sin\left(\frac{\pi}{6}\right)} - \frac{\cos\left(\frac{\pi}{6}\right)}{\sin\left(\frac{\pi}{6}\right)}\right|$

$= \ln|1-0| - \ln|2-\sqrt{3}|$

$= -\ln|2-\sqrt{3}|$

359. $\int_{\pi/4}^{\pi/3} \cot x\,dx$

$u = \sin(x),\ du = \cos(x)\,dx$

$= \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\cos(x)}{\sin(x)}\,dx = \int_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} \frac{du}{u}$

$= \left[\ln|u|\right]_{\frac{\sqrt{2}}{2}}^{\frac{\sqrt{3}}{2}} = \ln\left(\frac{\sqrt{3}}{2}\right) - \ln\left(\frac{\sqrt{2}}{2}\right)$

In the following exercises, integrate using the indicated substitution.

360. $\int \frac{x}{x-100}dx; u = x - 100$

$du = dx, \quad x = u + 100$

$= \int \frac{u+100}{u}du = \int \frac{u}{u} + \frac{100}{u}du$

$= \int \left(1 + \frac{100}{u}\right)du$

$= u + 100\ln|u| + C$

361. $\int \frac{y-1}{y+1}dy; u = y + 1$

$du = dy, \quad y - 1 = (y+1) - 2 = u - 2$

$\int \frac{u-2}{u}du$

$= \int \frac{u}{u} - \frac{2}{u}du = \int \left(1 - \frac{2}{u}\right)du$

$= u + 2\ln|u| + C$

362. $\int \frac{1-x^2}{3x-x^3}dx; u = 3x - x^3$

$du = 3 - 3x^2dx; \quad \frac{du}{3} = 1 - x^2dx$

$\int \frac{1}{3}\frac{du}{u} = \frac{1}{3}\ln|u| + C$

$= \frac{1}{3}\ln|3x - x^3| + C$

363. $\int \frac{\sin x + \cos x}{\sin x - \cos x}dx; u = \sin x - \cos x$

$du = (\cos(x) + \sin(x))dx$

$= \int \frac{du}{u} = \ln|u| + C = \ln|\sin(x) - \cos(x)| + C$